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ZETA-FUNCTION REGULARIZATION AND MULTI-INSTANTON DETERMINANTS *

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The determinant of the covariant laplacian is discussed using the ζ-function regularization procedure. For a self-dual solution to a general euclidean Yang-Mills gauge theory, an integral expression for its variation, with respect to the parameters of the solution, is obtained in terms of the general construction of Atiyah et al. The conformal properties are analyzed and an explicit expression is obtained for a part of the determinant containing all the non-trivial behavior. The remaining conformally invariant part has not been determined. The removal of the infrared divergences by considering the problem on a sphere is discussed in some detail.

1. Introduction

The complete characterization of the self-dual solutions to the euclidean Yang-Mills field equations, given by Atiyah, Drinfeld, Hitchin and Manin (ADHM) [1,2], has provided a framework within which quite explicit expressions can be constructed for certain Green functions, and for solutions of the massless Dirac equation, in the background field of these instantons [3–6]. Encouraged by these results, in this paper we shall consider the calculation of the determinants of operators defined using these background fields, in particular the determinant of the covariant laplacian.

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Such determinants would be expected to be relevant to the calculation of instanton contributions to functional integrals which serve to define euclidean quantum field theories [7]. However, if one accepts the validity of the dilute-gas approximation, the calculation of these determinants for self-dual or anti-self-dual multi-instanton solutions would have no direct physical relevance. There the dominant contributions to the functional integral come from field configurations with roughly equal numbers of instantons and anti-instantons which are only approximate solutions of the field equations [8]. But, in our view, the dilute-gas approximation is not obviously valid. It is very difficult to resolve questions of double counting with any degree of rigor, especially when the field configurations are not exact stationary points, let alone minima, of the action, Moreover, Atiyah and Jones [9] have shown that the space of self-dual instanton solutions exhausts much of the topological structure of the full space of gauge field configurations. One might even entertain the bold conjecture that the functional integrals occurring in quantum field theory could be well-approximated by gaussian integration about the strict minima of the action which are just given by multi-instanton or multi-anti-instanton solutions for arbitrary k. In the two-dimensional O(3) σ-model Frolov and Schwarz [10], and its CP^n generalizations, Berg and Lüscher [11] have completed the calculations corresponding to those we have considered here and have shown (at least in the O(3) case) that, as a contribution to the functional integral, the instantons behave as a Coulomb gas in its dense phase. If a similar result is shown for the Yang-Mills case, some of Witten's objections [12] to the dilute-gas approximation might be avoided without rendering physically irrelevant the beautiful mathematical results recently obtained concerning self-dual euclidean Yang-Mills fields.

As yet we have not been able to completely calculate any determinant, even for the simplest case of the covariant laplacian, D^2; thus the results presented here are preliminary. (Our calculations are undertaken in the fundamental representation of the gauge group but the methods of ref. [6] should permit results to be generalized to an arbitrary representation, at least in principle.) We shall use the ζ-function regularization procedure [13–15], because it seems the most natural generalization of the elementary definition of determinants to use in this context. A complete knowledge of the individual eigenvalues of the operator, which was employed by 't Hooft [7] and others [16] in the one-instanton case, is not necessary. We shall show that the variation of the determinant with respect to the instanton parameters can be expressed in terms of the ADHM construction in a manifestly gauge-invariant and finite manner. We employ the same basic approach as Brown and Creamer [17] in that our starting point is the relation

$$\delta \ln \det(-D^2) = \mathrm{Tr}[(D^2)^{-1}\delta D^2]\,, \tag{1.1}$$

suitably regularized. We use the ζ-function procedure to give a systematic derivation of their regularization prescriptions in sect. 2. (In eq. (1.1), $D_\mu = \partial_\mu + A_\mu$, where A is an antihermitian matrix formed from the gauge potentials.) Because euclidean space is non-compact $\det(-D^2)$ is infrared divergent. This divergence may be regularized by working on a sphere S^4 of radius a. This is considered in appendix A. But these divergences do not afflict $\delta \ln \det[-D^2]$, so it is sufficient to perform the analysis in flat space.

In sect. 3 we consider the evaluation of this expression for the variation of the determinant for the case of a self-dual gauge field. The ADHM construction [1,2] provides an algebraic construction of all self-dual solutions of the field equations, which is explicit except for the need to impose certain quadratic constraints on the parameters of the solution, and which is essentially independent of the gauge group. We obtain an explicit expression for $\delta \ln \det[-D^2]$ as an integral of a function depending on the parameters of the ADHM construction, generalizing results of Brown and Creamer [17]. The determinants for spin-$\frac{1}{2}$ and spin-1 (after the removal by projection of zero modes) are simply related to the scalar determinant [17,18], although they are not considered in this paper.

A crucial ingredient in obtaining this explicit expression is the short-distance expansion of a path-ordered exponential, taken along a straight-line path. Its derivation is outlined in appendix B.

Any regularization procedure will destroy conformal symmetry but should respect euclidean invariance, the analog of Poincaré invariance. On the other hand, the behavior under conformal transformations is calculable; Yoneya [19] has calculated it within the framework of dimensional regularization and Shwarz [20] has done the same for ζ-function regularization. In sect. 4 we review these results and use them to obtain a general expression for the transformation of $\det(-D^2)$ under a conformal transformation in terms of the ADHM construction, so that $\det(-D^2)$ may be expressed as a known conformally non-invariant part multiplied by a conformally invariant part which we have not determined. These results generalize those of Yoneya [19] and Schwarz [20].

The calculation of the conformal transformation property of $\det(-D^2)$ depends on an expression, with the ADHM framework, for the density of topological charge. This is derived in appendix C using the equations for the anomaly in the divergence of the axial current, which is itself also discussed.

The result we obtain for the conformally non-invariant part of $\det(-D^2)$ is expressed in terms of a conformally invariant matrix M which has previously occurred in constructing the Green function for the adjoint representation [6]. The reappearance of M is perhaps suggestive of deeper results. In appendix D we motivate its construction from considerations of conformal invariance alone.

Sect. 5 is devoted to evaluating the results of sect. 4 for the 't Hooft solutions [21], and comments are contained in sect. 6.

2. Zeta-function regularization for the covariant laplacian

If θ is a positive definite self-adjoint operator with eigenvalues $\lambda_1, \lambda_2, \ldots$ we can define an associated ζ-function, $\zeta_\theta(s)$, by

$$\zeta_\theta(s) = \sum_n \lambda_n^{-s} . \tag{2.1}$$

In the particular case where θ is a matrix acting on a finite-dimensional space U this sum is finite and, clearly

$$\zeta_\theta(0) = \dim \mathrm{U} \,, \tag{2.2}$$

$$\zeta_\theta'(0) = -\ln \det \theta \,. \tag{2.3}$$

Further, in this case, $\zeta_\theta(s)$ is everywhere an analytic function of s, and is given by the integral representation

$$\zeta_\theta(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \operatorname{Tr} \mathcal{G}(t)\, \mathrm{d}t \,, \tag{2.4}$$

where $\mathcal{G}(t) = \exp(-\theta t)$ and so may be defined by the equation,

$$\frac{\mathrm{d}\mathcal{G}}{\mathrm{d}t} = -\theta \mathcal{G} \,, \tag{2.5a}$$

subject to

$$\mathcal{G}(0) = 1 \,. \tag{2.5b}$$

For suitable partial differential operators $\mathcal{D}$, acting on infinite-dimensional function spaces, eqs. (2.2) and (2.3) may be used to provide (regularized) definitions of their dimension and determinant, and we may use eq. (2.4), together with the appropriate generalization of eq. (2.5):

$$\frac{\partial \mathcal{G}}{\partial t} = - \mathcal{D}\mathcal{G} \,, \tag{2.6a}$$

$$\mathcal{G}(x, y; 0) = \delta(x - y) \,, \tag{2.6b}$$

to calculate $\zeta_{\mathcal{D}}(s)$. In eq. (2.4),

$$\operatorname{Tr}\mathcal{G}(t) = \operatorname{Tr}[\mathrm{e}^{-t\mathcal{D}}] = \int \operatorname{tr}\ \mathcal{G}(x, x; t)\, \mathrm{d}^n x \,, \tag{2.7}$$

where tr denotes the operation of taking the trace over any internal (e.g., group) indices. Eqs. (2.6) define the heat kernel, whose properties are fully discussed by Gilkey [22] in the case where $\mathcal{D}$ is a second-order elliptic operator acting on a compact n-dimensional manifold. (See also refs. [23]). In this case the asymptotic properties of $\mathcal{G}(x, x; t)$ as $t \downarrow 0$ show that $\zeta_{\mathcal{D}}(s)$ is regular for $\mathrm{Re}(s) > \frac{1}{2}n$, there are poles at $s = 2$ and $s = 1$, but $\zeta_{\mathcal{D}}(s)$ is regular at $s = 0$.

Note that an immediate consequence of eq. (2.1) is that

$$\zeta_{\theta/\mu}(s) = \mu^s \zeta_\theta(s) \,, \tag{2.8}$$

for any scalar μ, which implies,

$$\zeta_{\theta/\mu}'(0) = \ln \mu\, \zeta_\theta(0) + \zeta_\theta'(0) \,. \tag{2.9}$$

Thus if we use eq. (2.3) to define the determinant of an operator,

$$\det(\mu^{-1}\theta) = \mu^{-\zeta_\theta(0)} \det \theta \,, \tag{2.10}$$

showing the approriateness of using $\zeta_\theta(0)$ as a generalization of dimension.

A problem with applying these results to the case we are interested in (the covariant laplacian, $-D^2$, on euclidean space R^4) is that this space is not compact. This gives rise to infrared divergences which would be regularized by considering the problem on a sphere S^4, where the results about the heat kernel [22] quoted above would apply rigorously. In appendix A we discuss the determinant of the operator $-D^2 + \frac{1}{6}R$ on a conformally flat space, with curvature scalar R, and deduce how, in the case of a sphere of radius a, the infrared divergences have been regularized. Since the part of the determinant which is divergent in the limit $a \to \infty$ is contained in a multiplicative factor independent of the potential A_μ, the variation of the logarithm of the determinant is finite in the flat-space case, which is obtained as $a \to \infty$. For this reason we may give a simple discussion of the calculation of $\delta \ln \det(-D^2)$ working in flat space and ignoring infrared problems.

To discuss the properties of $\zeta(s) \equiv \zeta_{-D^2}(s)$ we consider the asymptotic expansion of the heat kernel

$$\mathcal{G}(x, y; t) \sim \frac{1}{16\pi^2 t^2} \exp\left\{-\frac{1}{4t}(x-y)^2\right\} \sum_{n=0}^{\infty} a_n(x, y)\, t^n \,, \qquad \text{as } t \downarrow 0 \,. \tag{2.11}$$

Following DeWitt [24] and others [25], the coefficients a_n can be evaluated iteratively from eqs. (2.6), with $\mathcal{D} = -D^2$, by equating powers of t:

$$(x-y)_\mu D_\mu a_0(x; y) = 0 \,, \qquad a_0(x, x) = 1 \,, \tag{2.12a}$$

$$n a_n(x, y) + (x-y)_\mu D_\mu a_n(x, y) = D^2 a_{n-1}(x, y) \,, \qquad n \geqslant 1 \tag{2.12b}$$

Ignoring infrared problems, it is clear that the residues of $\zeta(s)$ at $s = 1,2$ and its value at $s = 0$ are controlled by the small-t behavior of $\mathcal{G}(x, x; t)$. Specifically

$$\mathrm{Res}_{s=2}\zeta(s) = \frac{1}{16\pi^2} \int \mathrm{tr}\, a_0(x, x)\, \mathrm{d}^4x \,, \tag{2.13a}$$

$$\mathrm{Res}_{s=1}\zeta(s) = \frac{1}{16\pi^2} \int \mathrm{tr}\, a_1(x, x)\, \mathrm{d}^4x \,, \tag{2.13b}$$

$$\zeta(0) = \frac{1}{16\pi^2} \int \mathrm{tr}\, a_2(x, x)\, \mathrm{d}^4x \,. \tag{2.13c}$$

Now eq. (2.12a) yields the path-ordered exponential

$$a_0(x, y) = \mathrm{P} \exp\{-\int_y^x A_\mu \mathrm{d}x_\mu\} \,, \tag{2.14}$$

taken along the straight-line path from x to y. Repeated differentiation of eq. (2.12b) yields

$$a_1(x, x) = [D^2 a_0(x, y)]_{x=y} = 0\,, \tag{2.15}$$

$$a_2(x, x) = [\tfrac{1}{6} D^2 D^2 a_0(x, y)]_{x=y} = \tfrac{1}{12} F_{\mu\nu} F_{\mu\nu}\,. \tag{2.16}$$

Thus the residue of $\zeta(s)$ at $s = 2$ is infrared divergent, whilst at $s = 1$ it vanishes, and

$$\zeta(0) = \frac{1}{12 \cdot 16\pi^2} \int d^4x \,\mathrm{tr}(F_{\mu\nu} F_{\mu\nu}) \tag{2.17}$$

$$= -\tfrac{1}{12}\, k\,, \tag{2.18}$$

for a self-dual solution, where the topological quantum number

$$k = -\frac{1}{16\pi^2} \int d^4x \,\mathrm{tr}(F_{\mu\nu} \,{}^*F_{\mu\nu}) \tag{2.19}$$

(the minus sign reflects the fact that $F_{\mu\nu}$ is antihermitian.)

To find the determinant we need to know $\zeta'(0)$. Using our knowledge of the residue of

$$\Gamma(s)\,\zeta(s) = \int_0^\infty dt\, t^{s-1}\, \mathrm{Tr}\,(e^{tD^2})\,. \tag{2.20}$$

at $s = 0$, and since

$$\mathrm{Tr}(e^{tD^2}) = \int d^4x \,\mathrm{tr}\,\mathcal{G}(x, x; t)\,,$$

we obtain the relation

$$\zeta'(0) - \frac{\gamma}{16\pi^2} \int d^4x \,\mathrm{tr}\, a_2(x, x) = \left[\int_0^\infty dt\, t^{s-1}\, \mathrm{Tr}(e^{tD^2}) - \frac{1}{16\pi^2 s} \int d^4x \,\mathrm{tr}\, a_2(x, x)\right]_{s=0}\,, \tag{2.21}$$

where the right-hand side has to be defined by analytic continuation and

$$\gamma = \frac{d}{ds}\left[\frac{1}{s\Gamma(s)}\right]_{s=0}$$

is Euler's constant. Apart from any difficulties of analytic continuation, it is not obvious how to evaluate eq. (2.21) without knowledge of the eigenvalues of $-D^2$. These difficulties are avoided if we consider $\delta\zeta(s)$, the variation in $\zeta(s)$ corresponding to a variation δA_μ in the potential A_μ. Since $a_0(x, x) = 1$, its variation is zero and $\delta\zeta(s)$ is regular at $s = 2, 1$. Further $\delta\zeta(0) = 0$, if the potential A_μ we are varying about defines a solution of the equations of motion, because the right-hand side of eq. (2.17) is proportional to the action. Hence

$$\delta\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \mathrm{d}t\, t^s\, \mathrm{Tr}[\mathrm{e}^{tD^2}\delta D^2] \ , \tag{2.22}$$

$$\delta\zeta'(0) = [\int_0^\infty \mathrm{d}t\, t^s \mathrm{Tr}[\mathrm{e}^{tD^2}\delta D^2]]_{s=0} \ , \tag{2.23}$$

where the integral is defined at $s = 0$ by analytic continuation from large values of $\mathrm{Re}(s)$. Integrating by parts,

$$\delta\zeta'(0) = [s \int_0^\infty \mathrm{d}t\, t^{s-1}\, \mathrm{Tr}[\mathrm{e}^{tD^2} G\, \delta D^2]]_{s=0} \ , \tag{2.24}$$

where $G(x, y)$ is the Green function,

$$D^2 G(x, y) = -\delta(x - y) \ . \tag{2.25}$$

Since

$$\delta D^2 = D_\mu \delta A_\mu + \delta A_\mu D_\mu \ , \tag{2.26}$$

we may rewrite eq. (2.24) as

$$\delta\zeta'(0) = [s \int_0^\infty \mathrm{d}t\, t^{s-1}\, \mathrm{Tr}[\mathrm{e}^{tD^2}\delta A_\mu \overrightarrow{D}_\mu G + \mathrm{e}^{tD^2} G \overleftarrow{D}_\mu \delta A_\mu]]_{s=0} \ , \tag{2.27}$$

using the notation,

$$\overrightarrow{D}_\mu G(x, y) = \frac{\partial}{\partial x^\mu} + A_\mu(x)\, G(x, y) \ , \tag{2.28a}$$

$$G(x, y)\overleftarrow{D}_\mu = -\frac{\partial}{\partial y^\mu} G(x, y) + G(x, y) A_\mu(y) \ . \tag{2.28b}$$

The advantage of eq. (2.24) or (2.27) over eq. (2.23) is clear: it leaves us to evaluate the residue of the pole at $s = 0$ in

$$\int \mathrm{d}t\, t^{s-1}\, \mathrm{Tr}[\delta A_\mu(\overrightarrow{D}_\mu G\, \mathrm{e}^{tD^2} + \mathrm{e}^{tD^2} G\overleftarrow{D}_\mu)] \ , \tag{2.29}$$

which is controlled by the small-t behavior. More precisely we need to find the constant term in the asymptotic expansion of

$$\int \mathrm{d}^4x\, \mathrm{d}^4y\, \mathrm{tr}[\delta A_\mu(x)(\overrightarrow{D}_\mu G(x, y)\, \mathcal{G}(y, x; t) + \mathcal{G}(x, y; t)\, G(y, x)\overleftarrow{D}_\mu)] \ , \tag{2.30}$$

as $t \downarrow 0$. If we attempted to do this by using

$$\mathcal{G}(x, y; t) = \Phi(x, y)\, \delta(x - y) + \mathrm{O}(t)\,, \qquad \text{as } t \downarrow 0\,, \tag{2.31}$$

where $\Phi(x, y) = a_0(x, y)$ is the path-ordered exponential of eq. (2.14), we would have the expression used by Brown and Creamer [17] as their starting point in a point-splitting approach and whose ambiguities they have to resolve by somewhat *ad hoc* means. On the other hand, the expression (2.30) automatically contains prescriptions for resolving these ambiguities leading to a justification of their procedures from the ζ-function method.

Following ref. [17] we note that

$$G(x, y) = \frac{1}{4\pi^2}\left(\frac{\Phi(x, y)}{(x - y)^2} + R(x, y)\right), \tag{2.32}$$

where $R(x, y)$ is non-singular at $x = y$. We shall now give a proof that only this regular part of $\mathcal{G}(x, y)$ contributes to the constant term in the asymptotic expansions of expression (2.30) as $t \downarrow 0$. This will be achieved if we show that

$$\frac{1}{4\pi^2}\int \mathrm{d}^4x\, \mathrm{d}^4y\, \mathrm{tr}\left[\delta A_\mu(x)\, \vec{D}_\mu \left\{\frac{\Phi(x, y)}{(x - y)^2}\right\} \mathcal{G}(y, x; t)\right] \tag{2.33}$$

has no constant term as $t \downarrow 0$. Now

$$\frac{1}{t^2}\int \mathrm{d}^4x\, x^{\mu_1} \ldots x^{\mu_N} (x^2)^{-M} \mathrm{e}^{-x^2/4t} \tag{2.34}$$

is finite and of order $t^{N/2-M}$, vanishing when N is odd, provided that $\frac{1}{2}N - M > -2$. So it suffices to consider, instead of (2.33),

$$\frac{1}{t^2}\int \mathrm{d}^4x\, \mathrm{d}^4y\, \mathrm{tr}\left[\delta A_\mu(x)\, \vec{D}_\mu \left\{\frac{\Phi(x, y)}{(x - y)^2}\right\} (a_0(y, x) + t a_1(y, x))\right] \mathrm{e}^{-(x-y)^2/4t}\,. \tag{2.35}$$

It is necessary to expand

$$\vec{D}_\mu \left\{\frac{\Phi(x, y)}{(x - y)^2}\right\} = (x - y)^{-2}\, \vec{D}_\mu \Phi(x, y) - 2(x - y)^{-4}(x - y)_\mu \Phi(x, y)\,, \tag{2.36}$$

multiplied by $a_0(y, x) + t a_1(y, x)$, in a Taylor series about $x = y$. To this end we need the results

$$\vec{D}_\mu \Phi(x, y) = \tfrac{1}{2}F_{\mu\nu}(x)\,(y - x)^\nu + \tfrac{1}{3}D_\lambda F_{\mu\nu}(x)\,(y - x)^\lambda (y - x)^\nu + \mathrm{O}(x - y)^2\,, \tag{2.37a}$$

$$\Phi(x, y)\, \Phi(y, x) = 1\,, \tag{2.37b}$$

$$a_1(y, x) = \mathrm{O}(x - y)^2\,, \tag{2.37c}$$

(Eqs. (2.37) can be established using $\partial_\mu(\Phi\psi) = \Phi\vec{D}_\mu\psi - \Phi\overleftarrow{D}_\mu\psi$ together with the techniques of ref. [24] or ref. [26] for evaluating derivatives of Φ.) Using (2.34) and

$$\frac{1}{16\pi^2 t^2}\int d^4x \frac{x^\mu x^\nu}{x^2}\, e^{-x^2/4t} = \delta^{\mu\nu}\,, \tag{2.38}$$

we see that the contribution of the singular part of $G(x, y)$ to (2.33) is

$$\frac{1}{48\pi^2}\int d^4x\, \mathrm{tr}[\delta A_\mu(x)\, D_\nu F_{\mu\nu}(x)]\,, \tag{2.39}$$

which vanishes if A_μ satisfies the equations of motion.

Thus only R contributes in eq. (2.27) and, using eq. (2.31) or

$$\int \frac{1}{16\pi^2 t^2}\, d^4x\, e^{-x^2/4t} = 1\,, \tag{2.40}$$

we deduce Brown and Creamer's expression [17]:

$$-\delta \ln \det(-D^2) = \delta\zeta'(0) = \int d^4x\, \mathrm{tr}[\delta A_\mu(x)\, J^\mu(x)]\,, \tag{2.41}$$

with

$$J_\mu(x) = \frac{1}{4\pi^2}\,[\overrightarrow{D}_\mu R(x, y) + R(y, x)\overleftarrow{D}_\mu]_{x=y}\,. \tag{2.42}$$

3. Evaluation of the determinant variation for self-dual fields

So far, in deriving eqs. (2.41) and (2.42), we have assumed no more than that A_μ satisfies the equations of motion. If we assume that the solution is self-dual, we can make these expressions more explicit within the framework of the ADHM construction. First, for the sake of completeness, we review those parts of the construction that we shall need, mainly using the notation of ref. [3]. We shall work with an Sp(n) group, where the formalism is most compact. The analysis may be easily adapted to any other gauge group.

Sp(n) may be defined as the group of $n \times n$ unitary matrices with quaternionic entries. (A quaternion may be thought of as a 2×2 matrix α of the form

$$\alpha = \alpha_0 - i\boldsymbol{\alpha}\cdot\boldsymbol{\sigma}\,, \tag{3.1}$$

where $\boldsymbol{\sigma}$ are the Pauli matrices and the α_μ are real.) Thus Sp(n) $\subset$ U($2n$). In the ADHM construction the potential $A_\mu(x)$ is given in the form

$$A_\mu(x) = v(x)^+ \partial_\mu v(x)\,, \tag{3.2}$$

where $v(x)$ is $(k + n) \times n$ matrix of quaternions defined by the equations

$$v(x)^+ v(x) = 1\,, \tag{3.3}$$

$$v(x)^+ \Delta(x) = 0\,. \tag{3.4}$$

In eq. (3.4), $\Delta(x)$ is a $(k+n) \times k$ matrix of quaternions, linear in the position variable x, having the structure

$$\Delta_{\lambda i}(x) = a_{\lambda i} + b_{\lambda i} x\,, \qquad 1 \leqslant i \leqslant k\,,$$
$$1 \leqslant \lambda \leqslant n+k\,, \tag{3.5}$$

where a, b are $(k+n) \times k$ matrices of quaternions and $x = x_0 - i\boldsymbol{x} \cdot \boldsymbol{\sigma}$. $\Delta(x)$ may be thought of as a map from a $2k$-complex dimensional space W to a $2(n+k)$-complex dimensional space V. Eqs. (3.3) and (3.4) are sufficient to ensure that

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \tag{3.6}$$

satisfies the self-duality equation

$$F_{\mu\nu} = {}^*F_{\mu\nu} \equiv \tfrac{1}{2}\epsilon_{\mu\nu\lambda\rho} F_{\lambda\rho}\,, \tag{3.7}$$

provided that a, b satisfy certain quadratic conditions, namely that a^+a, b^+b, and a^+b are all symmetric as $k \times k$ matrices of quaternions. This is equivalent to the condition that $\Delta(x)^+\Delta(x)$ viewed as a $k \times k$ quaternionic matrix is actually real for all x. To ensure that the solution is non-singular $\Delta(x)^+\Delta(x)$ must be non-singular for all x; b^+b must be non-singular for regularity at infinity.

Given such a pair of matrices a, b, eqs. (3.3) and (3.4) define A_μ up to gauge equivalence. Different pairs of matrices a, b may yield gauge-equivalent A_μ since eqs. (3.2) and (3.4) are invariant under

$$a \to QaK\,, \qquad b \to QbK\,, \tag{3.8}$$

$$v(x) \to Qv(x)\,, \qquad A_\mu(x) \to A_\mu(x)\,, \tag{3.9}$$

where $Q \in \mathrm{Sp}(n+k)$ and $K \in \mathrm{GL}(k, R)$. It may be shown that all matrices a, b, which yield gauge-equivalent solutions A_μ, are obtained by such transformations [1]. Thus we may hope to express any gauge-invariant functional of A_μ in terms of a and b,but it must be invariant under the transformation (3.8).

The topological quantum number of the solution, as given by eq. (2.19), is k.

To evaluate $J_\mu(x)$, as given by eq. (2.42), and hence the variation of the determinant, we need some information about $G(x,y)$ and $\Phi(x,y)$. Fortunately, there is a simple and elegant expression for the Green function within the framework of the ADHM construction [3,4]:

$$G(x,y) = \frac{1}{4\pi^2}\,\frac{v(x)^+ v(y)}{(x-y)^2}\,. \tag{3.10}$$

We do not know of any comparably simple form for $\Phi(x,y)$ but it is not difficult to arrive at the following ansatz on the basis of gauge covariance and euclidean transformation properties:

$$\Phi(x,y) = v(x)^+ \{1 + (x-y)^2\, bH(x,y)\, b^+\}\, v(y)\,. \tag{3.11}$$

Using the defining equations for $\Phi(x, y)$,

$$(x - y)_\mu \overrightarrow{D}_\mu \Phi(x, y) = 0 = \Phi(x, y) \overleftarrow{D}_\mu (x - y)_\mu \,, \tag{3.12}$$

we may calculate a power-series expansion of $H(x, y)$ about $x = y$,

$$H(x, y) = \tfrac{1}{2} f(\bar{x}) + \tfrac{1}{12} f(\bar{x}) [(x - y)\, \Delta(\bar{x})^+ b - b^+ \Delta(\bar{x})\, (x - y)^+]\, f(\bar{x}) + \mathrm{O}(|x - y|^2) \tag{3.13}$$

Here $\bar{x} = \frac{1}{2}(x + y)$ and

$$f(x) = [\Delta(x)^+ \Delta(x)]^{-1} . \tag{3.14}$$

Further details are given in appendix B.

Using eqs. (3.11) and (3.13), it is straightforward to show that, as defined by eq. (2.42),

$$J_\mu = \frac{1}{12\pi^2}\, v^+ b f (e_\mu \Delta^+ b - b^+ \Delta e_\mu^+) f b^+ v \,, \tag{3.15}$$

where $e_\mu = \partial_\mu x$.

We now consider δA_μ. In sect. 2 it was arbitrary, but we now wish to examine the consequences of taking it to correspond to a variation of A_μ within the space of solutions. This is the kind of variation we need to integrate to obtain an expression for $\det(-D^2)$ itself. Consider varying Δ by an amount $\delta\Delta = \delta a + \delta b x$. Then v will have to be changed to $v + \delta v$ where

$$\delta v^+ \Delta + v^+ \delta\Delta = 0 \,, \tag{3.16}$$

and

$$\delta v^+ v + v^+ \delta v = 0 \,. \tag{3.17}$$

The general solution to eq. (3.16) is of the form

$$\delta v^+ = -v^+ \delta\Delta f \Delta^+ + \delta n^+ v^+ \,, \tag{3.18}$$

and eq. (3.17) implies

$$\delta n + \delta n^+ = 0 \,. \tag{3.19}$$

Correspondingly

$$\begin{aligned} \delta A_\mu &= v^+ \partial_\mu \delta v + \delta v^+ \partial_\mu v \\ &= -v^+ \delta\Delta f \Delta^+ \partial_\mu v - v^+ \partial_\mu \Delta f \delta\Delta^+ v - \delta n A_\mu + A_\mu \delta n + \partial_\mu \delta n \qquad (3.20) \\ &= v^+ (\delta\Delta f \partial_\mu \Delta^+ - \partial_\mu \Delta f \delta\Delta^+)\, v + D_\mu \delta n \,. \qquad (3.21) \end{aligned}$$

The term $D_\mu \delta n = \partial_\mu \delta n + [A_\mu, \delta n]$ corresponds to an infinitesimal gauge transformation. The conservation equation

$$D_\mu J_\mu = 0 \,, \tag{3.22}$$

which may be directly verified using eq. (3.15) and the properties of Δ, etc., ensures that it makes no contribution in eq. (2.41).

A central rôle in the ADHM construction is played by the gauge-invariant projection operator

$$P(x) = v(x)\, v(x)^+ \tag{3.23}$$

$$= 1 - \Delta f \Delta^+ \,, \tag{3.24}$$

which satisfies $P^2 = P$ and $P^+ = P$. We may rewrite the integrand of eq. (2.41) in terms of P and δP as follows:

$$12\pi^2 \ \mathrm{tr}(J_\mu \delta A_\mu) = \mathrm{tr}[Pbf(e_\mu \Delta^+ b - b^+ \Delta e_\mu^+) f b^+ P(\delta\Delta f \partial_\mu \Delta^+ - \partial_\mu \Delta f \delta \Delta^+)] \,. \tag{3.25}$$

Now since

$$P\delta P = -P\delta\Delta f \Delta^+ \,, \tag{3.26}$$

and

$$P\partial_\mu(P\partial_\mu P) = -4Pbfb^+ \,, \tag{3.27}$$

$$48\pi^2 \mathrm{tr}(J_\mu \delta A_\mu) = \mathrm{tr}\{P[\delta P, \partial_\mu P]\, P(\partial_\nu(P\partial_\nu P)\, \partial_\mu P - \partial_\mu P\, \partial_\nu(\partial_\nu P P))\} \tag{3.28}$$

$$= \tfrac{1}{2}\, \mathrm{tr}\{\delta P(\partial^2 P(1-P)\, \partial^2 P - 2\partial_\mu P\, \partial^2 P\, \partial_\mu P)\} \,, \tag{3.29}$$

where, to obtain the last expression, we have used

$$P\partial_\mu P P = 0 \tag{3.30}$$

and other relations following from $P^2 = P$.

We have sought unsuccessfully to write eq. (3.29) in the form

$$\mathrm{tr}(J_\mu \delta A_\mu) = \delta \ \mathrm{tr}\, B + \partial_\mu \ \mathrm{tr}\ C_\mu \,, \tag{3.31}$$

where the second term would give no contribution in eq. (2.41) on integrating. Ideally one would like to express the integral of tr(B) as a function of a, b, which would have to be invariant under the transformations (3.8). So far all we have been able to establish about this integral is its conformal transformation properties to which we address ourselves in sects. 4, 5.

4. Conformal transformations

Conformal transofrmations on four-dimensional euclidean space are conveniently expressed in terms of quaternions. The transformations take the form $x \to x'$ with

$$x' = (\alpha x + \beta)(\gamma x + \chi)^{-1} \,, \tag{4.1}$$

where $\alpha, \beta, \gamma, \chi$ are quaternions such that

$$\kappa^2 = \det\begin{pmatrix} \alpha & \beta \\ \gamma & \chi \end{pmatrix} = |\alpha\gamma^{-1}\chi\gamma - \beta\gamma|^2 \,, \tag{4.2}$$

which is inevitably positive, is non-zero. (Scaling α, β, γ, χ by a real number has no effect on eq. (4.1) so that κ may be fixed at unity if so desired.) It is straightforward to show from eq. (4.1) that

$$(\mathrm{d}x')^2 = (\Omega(x))^2\,(\mathrm{d}x)^2\,, \tag{4.3}$$

with

$$\Omega(x) = \kappa\,|\gamma x + \chi|^{-2}\,, \tag{4.4}$$

exhibiting the conformal property of the transformation (4.1).

The conformal invariance of the Yang-Mills equations amounts to the statement that if $A_\mu(x)$ satisfies the field equations so does

$$A'_\rho(x) = A_\mu(x')\,\frac{\partial x'_\mu}{\partial x_\rho}\,, \tag{4.5}$$

where on the right-hand side of eq. (4.5) x' is defined as a function of x by eq. (4.1). The effects of this transformation on D_μ and D^2 are described by

$$\frac{\partial}{\partial x'_\nu} + A_\nu(x') = \frac{\partial x_\mu}{\partial x'_\nu}\left(\frac{\partial}{\partial x_\mu} + A'_\mu(x)\right)\,, \tag{4.6}$$

and

$$\left(\frac{\partial}{\partial x'_\nu} + A_\nu(x')\right)^2 \phi'(x') = (\Omega(x))^{-3}\left(\frac{\partial}{\partial x_\nu} + A'_\nu(x)\right)^2 \phi(x)\,, \tag{4.7}$$

where $\phi'(x') = \Omega(x)^{-1}\phi(x)$. Now $\Omega(x) = \omega(x')^{-1}$ with

$$\omega(x') = \kappa\,|x'\gamma - \alpha|^{-2}\,, \tag{4.8}$$

So the determinant of $(\omega^{-3}D^2\omega)$, where ω is defined by replacing x' by x in eq. (4.8), equals the determinant of D'^2, where

$$D'_\nu = \frac{\partial}{\partial x_\nu} + A'_\nu(x)\,, \tag{4.9}$$

since eq. (4.7) makes it clear that these operators have equal eigenvalues.

These results enable us to calculate the change in $\zeta(s)$ corresponding to an infinitesimal conformal transformation $A_\mu \to A'_\mu$ [20], by using eq. (2.22) with $\delta D^2 = \delta(\omega^{-3}D^2\omega)$,

$$\delta\zeta(s) = \frac{1}{\Gamma(s)}\int_0^\infty \mathrm{d}t\; t^s\; \mathrm{Tr}\,[\mathrm{e}^{tD^2}(D^2\delta\omega - 3\delta\omega D^2)] \tag{4.10}$$

$$= \frac{2}{\Gamma(s)}\int_0^\infty \mathrm{d}t\; t^s\; \mathrm{Tr}[\mathrm{e}^{tD^2}D^2\delta\omega] \tag{4.11}$$

$$= \frac{2s}{\Gamma(s)}\int_0^\infty \mathrm{d}t\; t^{s-1}\; \mathrm{Tr}[\mathrm{e}^{tD^2}\delta\omega]\,. \tag{4.12}$$

Differentiating we see that $\delta\zeta'(0)$ is given by the residue of the pole in

$$2\int_0^\infty \mathrm{d}t\, t^{s-1}\, \mathrm{tr}[\mathcal{G}(x,x;t)\,\delta\omega(x)]\,, \tag{4.13}$$

at $s = 0$. Thus

$$\delta\zeta'(0) = \frac{1}{8\pi^2}\int \mathrm{tr}\, a_2(x,x)\,\delta\omega(x)\,\mathrm{d}^4x \tag{4.14}$$

$$= \frac{1}{96\pi^2}\int \mathrm{tr}(F_{\mu\nu}F_{\mu\nu})\,\delta\omega(x)\,\mathrm{d}^4x \tag{4.15}$$

We shall now apply this result to self-dual solutions, beginning by calculating the change in a, b corresponding to the transformation $A_\mu \to A'_\mu$. If $A_\mu = v^+\partial_\mu v$, we see that A'_μ is given similarly in terms of v' where

$$v'(x) = v(x'(x))\,. \tag{4.16}$$

Then

$$v'^+(x)\,\Delta'(x) = 0\,, \tag{4.17}$$

if

$$\Delta'(x) = a' + b'x\,, \tag{4.18}$$

with

$$a' = a\chi + b\beta\,, \tag{4.19a}$$

$$b' = a\gamma + b\alpha\,. \tag{4.19b}$$

Consider the infinitesimal conformal transformation given by taking $\alpha, \beta, \gamma, \chi$ to be $1 + \delta\alpha$, $\delta\beta$, $\delta\gamma$, $1 + \delta\chi$. Then

$$\delta\omega = \tfrac{1}{2}\,\mathrm{tr}(\delta\chi - \delta\alpha) + \mathrm{tr}(x\delta\gamma) \tag{4.20}$$

$$\equiv \sigma + \tau_\mu x_\mu\,. \tag{4.21}$$

The key step in evaluating the right-hand side of eq. (4.15) is the observation that

$$\tfrac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F_{\mu\nu}) = \partial^2 T\,, \tag{4.22}$$

where T is given in terms of the ADHM construction by

$$T = \mathrm{tr}_W\,[b^+Pbf + b^+bf]\,, \tag{4.23}$$

where the trace is over the $2k$-dimensional space W. This relation was obtained by considering the equation relating the divergence of the expectation value of the axial vector current to the Adler-Bell-Jackiw anomaly, and its derivation is outlined in appendix C. The value of eq. (4.22) is immediate; it enables us to express the integral in eq. (4.15) as a surface integral at infinity:

$$\frac{1}{96\pi^2}\int \mathrm{tr}(F_{\mu\nu}F_{\mu\nu})\,\delta\omega\,\mathrm{d}^4x = \frac{1}{48\pi^2}\int \partial^2 T(\sigma+\tau_\mu\chi_\mu)\,\mathrm{d}^4x \tag{4.24}$$

$$= \frac{1}{48\pi^2}\lim_{r\to\infty}\int\{\partial_\nu T(\sigma+\tau_\mu x_\mu) - T\tau_\nu\}\,\mathrm{d}S_\nu\,, \tag{4.25}$$

where the integral is taken over the sphere $|x| = r$. The term in T involving b^+Pbf decreases faster than $|x|^{-4}$ at infinity and so makes no contribution in eq. (4.25). The asymptotic form of the other term is given by

$$b^+bf = |x|^{-2} - (x^+b^+a + a^+bx)(b^+b)^{-1}|x|^{-4} + \mathrm{O}(|x|^{-4})\,, \tag{4.26}$$

which yields in eq. (4.25),

$$\delta\zeta'(0) = \tfrac{1}{24}\,\mathrm{tr_W}\,[(b^+a\,\tau + \tau^+a^+b)(b^+b)^{-1}] - \tfrac{1}{6}k\sigma \tag{4.27}$$

$$= \tfrac{1}{12}\,\mathrm{tr_W}\,[(b^+a\,\delta\gamma + \delta\gamma^+a^+b)(b^+b)^{-1}] - \tfrac{1}{12}k\,\mathrm{tr}(\delta\chi - \delta\alpha) \tag{4.28}$$

$$= \tfrac{1}{12}\,\mathrm{tr_W}\,[\delta(b^+b)(b^+b)^{-1}] - \tfrac{1}{12}k\,\mathrm{tr}(\delta\chi + \delta\alpha) \tag{4.29}$$

$$= \tfrac{1}{12}\,\delta\,\ln\,[\det(b^+b)^2\kappa^{-2k}]\,, \tag{4.30}$$

where b^+b is regarded as a $k\times k$ real matrix and we have used eq. (4.2).

Thus, in order to construct a function having the same conformal transformation properties as $\zeta'(0)$ we need to find a function $\lambda(a, b)$ satisfying

$$\lambda(QaK, QbK) = (\det K)^{-4}\lambda(a, b)\,, \tag{4.31}$$

for $Q\in \mathrm{Sp}(n+k)$, $K\in \mathrm{GL}(k, R)$ and

$$\lambda(a', b') = \kappa^{2k}\lambda(a, b)\,, \tag{4.32}$$

with a', b' related to a, b by the conformal transformation of eqs. (4.19).

Matrices with simple conformal transformation properties were introduced in ref. [6] in the course of constructing Green functions for adjoint and higher representations. In particular, for the adjoint representation, the matrix M defined by

$$\begin{aligned}(M^{-1})_{ij,i'j'} &= (a^+a)_{ii'}(b^+b)_{jj'} + (b^+b)_{ii'}(a^+a)_{jj'}\\ &\quad - \mathrm{tr}_2\,[(a^+b)_{ii'}(b^+a)_{j'j}]\,,\end{aligned} \tag{4.33}$$

where the trace is over the two-dimensional quaternionic labels, satisfies

$$M^{-1}\to(K^{\mathrm T}\otimes K^{\mathrm T})M^{-1}(K\otimes K)\,, \tag{4.34}$$

under the transformation (3.8), and

$$M^{-1}\to\kappa^2M^{-1}\,, \tag{4.35}$$

under the conformal transformation $a \to a'$, $b \to b'$. Taking determinants, we obtain

$$\det M \to (\det K)^{-4k} \det M\,, \tag{4.36}$$

from eq. (4.34) and

$$\det M \to \kappa^{-2k^2} \det M\,, \tag{4.37}$$

from eq. (4.35). So

$$\lambda = (\det M)^{1/k} \tag{4.38}$$

has properties (4.31) and (4.32) and

$$\delta\zeta'(0) = \frac{1}{12k}\, \delta \ln\, [\det\{(b^+b \otimes b^+b)\, M\}]\ , \tag{4.39}$$

so that, with an explicit regularization scale μ,

$$\delta \det(-D^2/\mu^2) = \delta \det\{\mu^{-2}(b^+b \otimes b^+b)\, M\ \}^{-1/12k}\ , \tag{4.40}$$

for conformal transformations.

From eq. (4.39) we deduce

$$\frac{\det(-D^2/\mu^2)}{\det(-D_0^2/\mu^2)} = \det\{\mu^{-2}(b^+b \otimes b^+b)\, M\}^{-1/12k} C\,, \tag{4.41}$$

where C is a conformal invariant and D_0^2 is obtained from D^2 by putting $A_\mu = 0$.

5. Evaluation for 't Hooft solutions

In this section we evaluate the results of sect. 4 for the 't Hooft solutions. In terms of the ADHM construction these solutions correspond to

$$a_{0i} = y_0\lambda_i\,, \qquad b_{0i} = -\lambda_i\,, \qquad 1 \leqslant i \leqslant k\,, \tag{5.1a}$$

$$a_{ij} = y_i\lambda_0\delta_{ij}\,, \qquad b_{ij} = -\lambda_0\delta_{ij}\,, \qquad 1 \leqslant i, j \leqslant k\,, \tag{5.1b}$$

where $\lambda_0, \lambda_1, ..., \lambda_k$ are the scales and $y_0, y_1, y_2, ...\, y_k$ the quaternionic representation of the "positions" of the instantons [3,5]. From these equations it follows that

$$(b^+b)_{ij} = \lambda_0^2\delta_{ij} + \lambda_i\lambda_j\,, \tag{5.2}$$

and

$$(M^{-1})_{ij,rs} = \lambda_0^4\lambda_i\lambda_j\lambda_r\lambda_s[s_{0i}\delta_{ir} + s_{ij}\delta_{ir}\delta_{js} + s_{0j}\delta_{js}]\ , \tag{5.3}$$

where

$$s_{ij} = (y_i - y_j)^2\lambda_j^{-2}\lambda_i^{-2}\ . \tag{5.4}$$

We wish to calculate $(\det b^+b)^{2k} \det M$. We know this is unchanged if the same permutation on $k+1$ objects is applied simultaneously to $(\lambda_0, \lambda_1, ..., \lambda_k)$ and $(y_0, y_1, ..., y_k)$, because the $(k+1)$ points occur on an equal footing in the solution. This symmetry is not manifest, being a consequence of the invariance under $a \to aK$, $b \to bK$, $K \in \mathrm{GL}(k,R)$

It is easy to calculate $\det b^+b$ and one obtains

$$\det(b^+b)^{2k} = \lambda_0^{4k(k-1)} (\sum_{i=0}^{k} \lambda_i^2)^{2k} . \tag{5.5}$$

On the other hand $\det M$ is more difficult. If

$$S_{ij,rs} = s_{0i}\delta_{ir} + s_{0j}\delta_{js} + s_{ij}\delta_{ir}\delta_{js} , \tag{5.6}$$

$$\det M^{-1} = \lambda_0^{4k(k-1)} (\prod_{i=0}^{k} \lambda_i^2)^{2k} \det S . \tag{5.7}$$

To evaluate $\det S$ we use

$$\delta \ln \det S = \mathrm{tr}(S^{-1}\delta S) \tag{5.8}$$

Using the methods of Brown et al. [27], we obtain

$$\begin{aligned} S^{-1}_{ij,rs} &= t_{ij}\delta_{ir}\delta_{js} - \tfrac{1}{2}t_{ij}\delta_{rs}(\delta_{jr} + \delta_{ir}) - \tfrac{1}{2}t_{rs}\delta_{ij}(\delta_{is} + \delta_{ir}) \\ &+ \tfrac{1}{2}\delta_{ij}\delta_{rs}t_{ir} + \tfrac{1}{2}\delta_{ij}\delta_{rs}\delta_{ir} \sum_{l=0}^{k} t_{il} \\ &- \tfrac{1}{2}t_{ij}t_{rs}(m_{ir} + m_{js} - m_{is} - m_{jr}) , \end{aligned} \tag{5.9}$$

where

$$\begin{aligned} t_{ij} &= S^{-1}_{ij} , \qquad \text{if } i \neq j , \\ &= 0 , \qquad \text{if } i = j , \qquad 0 \leqslant i, j \leqslant k , \end{aligned} \tag{5.10}$$

and m is the matrix inverse of

$$p_{ij} = \delta_{ij} \sum_{l=0}^{k} t_{il} - t_{ij} , \qquad 1 \leqslant i,j \leqslant k . \tag{5.11}$$

From eqs. (5.6) and (5.9)–(5.11) it follows that

$$\mathrm{tr}(S^{-1}\delta S) = 2 \sum_{0 \leqslant ij, \leqslant k} s^{-1}_{ij}\, \delta s_{ij} + \sum_{1 \leqslant i,j \leqslant k} m_{ij}\delta p_{ij} , \tag{5.12}$$

so

$$\det S = 2^k \det p \prod_{0 \leqslant i < j \leqslant k} s_{ij}^2 , \tag{5.13}$$

where the constant of proportionality 2^k can be evaluated by comparing the coefficients of

$$\prod_{0\leqslant i<j\leqslant k} s_{ij} \prod_{0\leqslant i<j\leqslant k} s_{ij} \,. \tag{5.14}$$

(Such terms come from the main diagonals of det S and det p.)

It remains to evaluate det p. It follows from eqs. (5.5), (5.7) and (5.13) that det p possesses the permutation symmetry on $k + 1$ objects possessed by (det $b^+b)^{2k}$ det M. (Although this is not manifest in its definition (5.11), it can be established from it by row and column operations.) Clearly det p consists of sums of products of t_{ij}'s each term in the sum having k factors. All the terms involving t_{i0}, for some i, come from the main diagonal of p, and all distinct products of this sort that occur do so just once and with coefficient plus 1. By the permutation symmetry, all of the products occurring have coefficient plus 1. It may be shown inductively that there are $(k+ 1)^{k-1}$ terms in det p, and that these terms are just products of k t_{ij}'s,

$$t_{i_1j_1}t_{i_2j_2} \,....\, t_{i_kj_k} \,, \tag{5.15}$$

in which (i) each index form 0 to k occurs at least once as a suffix, and (ii) there is no factor which can be written in the cyclic form

$$t_{i_1i_2}t_{i_2i_3} \cdots t_{i_ri_1} \,. \tag{5.16}$$

This rule for evaluating det p was first given by Sylvester [28]; det p is called Sylvester's unisignant. The symmetric case of interest here is associated with the name of Borchardt [29], who presented a detailed derivation of the above rule.

Thus we have obtained

$$\det M^{-1} (\det b^+b)^{-2k} = 2^k(\sum_{i=0}^{k} \lambda_i^2)^{-2k}(\prod_{i=0}^{k} \lambda_i^2)^{2k} \prod_{0\leqslant i<j\leqslant k} s_{ij}^2 \det p \,, \tag{5.17}$$

with det p given by the above construction.

6. Comments and conclusions

Whilst we have not succeeded in completely computing det $(-D^2)$, we feel that the approach developed in sects. 2 and 3 may contain the elements of a solution, and that the conformally non-invariant piece, which we have isolated, contains an important part of the structure of the final result. There remains the problem of integrating the variation δA_μ, which has been solved for the 't Hooft solutions by Brown and Creamer [17], and furthermore evaluating the spatial integral, which has yet to be done for the 't Hooft case.

The remaining part of the determinant must be a function of the conformal invariants formed from the instanton parameters also invariant under any symmetries of the general k-instanton gauge potential. Since there is a unique correspondence between the gauge-equivalence class of the potential and the matrices a and b, up to a transformation of the form of (3.8), it is sufficient to find functions of a and b invariant under eq. (3.8) and the conformal transformations (4.19). We have presented one such invariant in appendix D. Another approach to the construction of such invariants is to consider the matrix $\mathcal{M}$ which occurs in the construction of the Green function for D^2 in the adjoint representation. This matrix is a conformally invariant function of the matrices a, b determining the solution, and

$$\mathcal{M} \to (Q \otimes Q)\mathcal{M}(Q^+ \otimes Q^+)\,, \tag{6.1}$$

under the transformation (3.8). Since $Q \in \mathrm{Sp}(k+n)$, the eigenvalues of $\mathcal{M}$ are both conformally invariant and unchanged by (3.8). It is unclear how many independent invariants there are. If this number were small and independent of k, it would give hope that the determinant could be completely evaluated.

If the determinant could be completely evaluated, it would be fairly straightforward to extend the analysis to derive the multi-instanton contributions to the complete gauge theory partition functions. We may then be able to discuss the statistical mechanics of instantons and answer questions of physical interest, such as the validity of the dilute-gas approximation and, perhaps, deeper questions about the structure of gauge theories.

After completing this stage of our study we received a paper by Belavin, Fateev, Schwarz and Tyupkin [34], whose results have some overlap with ours.

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Appendix A

Conformally flat spaces

So far we have only discussed determinants of differential operators acting on functions defined on flat euclidean space R^4. The non-compactness of euclidean space introduces divergences in the expressions for $\zeta(s)$, but these disappear on a noncompact space such as the sphere S^4. In studying what changes are introduced by working on a curved space we begin by considering the effect of a conformal scaling of the metric.

The conformal scalar operator $\mathcal{D}$ is defined by

$$\mathcal{D}\phi = -D^2\phi + \tfrac{1}{6}R\phi \tag{A.1}$$

$$= -g^{-1/2}D_\mu(g^{1/2}g^{\mu\nu}D_\nu\phi) + \tfrac{1}{6}R\phi\,, \tag{A.2}$$

where R is the curvature scalar. (In eq. (A.1) $D^2 = g^{\mu\nu}D_\mu D_\nu$, where D_μ is the covariant derivative containing both the Yang-Mills potential A_μ and the connection associated with the metric tensor $g_{\mu\nu}$, but in calculating the equivalent form in eq. (A.2) it is only necessary to take $D_\mu = \partial_\mu + A_\mu$.) If $\mathcal{D}'$ is similarly defined by the metric

$$g'_{\mu\nu}(x) = (\omega(x))^2\, g_{\mu\nu}(x)\,, \tag{A.3}$$

we have the conformal transformation property

$$\mathcal{D}' = \omega^{-3}\,\mathcal{D}\omega\,. \tag{A.4}$$

Then, as in eq. (4.13), the variation $\delta\zeta'_{\mathcal{D}}(0)$, corresponding to an infinitesimal transformation given by $\omega = 1 + \delta\omega$, equals the residue at $S = 0$ of

$$2\int_0^\infty \mathrm{d}t\, t^{s-1}\, \mathrm{Tr}(\mathrm{e}^{-t\mathcal{D}}\delta\omega) \tag{A.5}$$

Now,

$$\mathrm{Tr}(\mathrm{e}^{-t\mathcal{D}}\delta\omega) = \int \mathrm{d}^4x\, \mathrm{tr}(\mathcal{G}(x, x; t)\, \delta\omega(x)\,. \tag{A.6}$$

The asymptotic expansion of the heat kernel, which for the diagonal elements takes the form

$$\mathcal{G}(x, x; t) = \frac{g(x)^{1/2}}{16\pi^2 t^2} \sum_{n=0}^{\infty} a_n(x, x)\, t^n\,, \qquad \text{as } t \downarrow 0\,, \tag{A.7}$$

has been discussed by many authors [24,25]. Only the a_2 term contributes to the residue at $S = 0$ of (A.5) so that [20]

$$\delta\zeta'_{\mathcal{D}}(0) = \frac{1}{8\pi^2} \int \mathrm{d}^4x\, g(x)^{1/2}\, \mathrm{tr}\, a_2(x, x)\, \delta\omega(x)\,. \tag{A.8}$$

It can be shown that [24,25]

$$a_2(x, x) = \tfrac{1}{12} F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{180}(R^{\mu\nu\sigma\tau}R_{\mu\nu\sigma\tau} - R^{\mu\nu}R_{\mu\nu} + \nabla^2 R)\,, \tag{A.9}$$

where $\nabla^2 R = g^{-1/2}\partial_\mu(g^{1/2}g^{\mu\nu}\partial_\nu R)$.

For a conformally flat space,

$$g_{\alpha\beta}(x) = (\Omega(x))^2 \delta_{\alpha\beta}\,, \tag{A.10}$$

the vanishing of the Weyl tensor implies

$$R^{\mu\nu\sigma\tau}R_{\mu\nu\sigma\tau} - 2R^{\mu\nu}R_{\mu\nu} + \tfrac{1}{3}R^2 = 0\,. \tag{A.11}$$

In particular, for a sphere S^4 of radius a

$$\Omega(x) = 2(1 + x^2/a^2)^{-1}\,, \tag{A.12}$$

for coordinates obtained by stereographic projection. Then

$$R = \frac{12}{a^2}\,, \qquad R_{\mu\nu} = \frac{3}{a^2}\, g_{\mu\nu} \tag{A.13}$$

For a change from a to $a + \delta a$,

$$\delta\omega = \delta\Omega/\Omega\,, \tag{A.14}$$

and

$$\delta\zeta'_{\mathcal{D}}(0) = \frac{1}{32\pi^2} \int d^4x\, \delta(\Omega^4) \left\{ \tfrac{1}{12}\, \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) - \frac{1}{15}\frac{N}{a^4} \right\}, \tag{A.15}$$

where N is the dimension of the gauge field representation. Now, given the metric (A.10)

$$\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = \Omega^{-4}\, \mathrm{tr}(F_{\mu\nu}F_{\mu\nu})\,, \tag{A.16}$$

and $F_{\mu\nu}$ is conformally invariant under the change of metric of eq. (A.3). Thus

$$\delta\zeta'_{\mathcal{D}}(0) = \delta\left[\frac{1}{96\pi^2} \int d^4x\, \ln\Omega\, \mathrm{tr}(F_{\mu\nu}F_{\mu\nu}) - \tfrac{1}{45} N \ln a \right], \tag{A.17}$$

In eq. (A.17) the integral is convergent, at least for solutions of the field equations, and gives a zero contribution in the flat space limit $a \to \infty$. The term involving $\ln a$ diverges as $a \to \infty$, but this divergence cancels if we consider

$$\det \mathcal{D} / \det \mathcal{D}_0\,, \tag{A.18}$$

where $\mathcal{D}_0$ is constructed by putting $A_\mu = 0$ in $\mathcal{D}$. We see no reason for the difficulties recently encountered in this context by Patrasciou [30]; eq. (A.17) provides an alternative justification for 't Hooft's method of changing the regularization scale from S^4 to flat space [7].

From the definition of $\zeta_{\mathcal{D}}(s)$, as in eq. (2.4), using eqs. (A.7) and (A.9) we obtain on S^4,

$$\zeta_{\mathcal{D}}(0) = -\tfrac{1}{12}k - \tfrac{1}{90}N\,. \tag{A.19}$$

Thus if we introduce a scale parameter μ, the dependence of $\det(\mathcal{D}/\mu^2)$ on μ and the divergent part of its dependence on a are both contained in a factor

$$\mu^{k/6}(\mu a)^{N/45}\,. \tag{A.20}$$

The factor $\mu^{k/6}$ is that which would be obtained by a scaling transformation on flat space using the analysis of sect. 4.

Appendix B

Expansion of $\Phi(x,y)$

In order to calculate $H(x, y)$ we start from the defining equations (3.12) for $\Phi(x,y)$ Substituting in from eq. (3.11), writing

$$h(x, y) = (x - y)^2\, H(x, y)\,, \tag{B.1}$$

yields

$$(x-y)_\mu \frac{\partial}{\partial x_\mu} h(x,y) = (x-y)^2 f(x) + (x-y) f(x)\,\Delta(x)^+ b h(x,y)\,, \tag{B.2}$$

$$(x-y)_\mu \frac{\partial}{\partial y_\mu} h(x,y) = -(x-y)^2 f(y) + h(x,y)\, b^+ \Delta(y) f(y)\,(x-y)^+\,. \tag{B.3}$$

Setting $\bar{x} = \frac{1}{2}(x+y)$, $\beta = \frac{1}{2}(x-y)$ and subtracting eq. (B.3) from eq. (B.2),

$$\beta_\mu \frac{\partial}{\partial \beta_\mu} h = 2\beta^2 \{f(\bar{x}+\beta) + f(\bar{x}-\beta)\}$$
$$+ \beta f(\bar{x}+\beta)\,\Delta(\bar{x}+\beta)^+ bh - hb^+\Delta(\bar{x}-\beta) f(\bar{x}-\beta)\,\beta^+\,, \tag{B.4}$$

which can be solved iteratively to yield a Taylor-series expansion for h in β_μ, starting from $h(x.x) = 0$. Thus we obtain

$$h = 2\beta^2 \{f(\bar{x}) + \tfrac{1}{3} f(\bar{x})[\beta\Delta(\bar{x})^+ b - b^+\Delta(\bar{x})\,\beta^+]\, f(\bar{x})\} + O(\beta^4) \tag{B.5}$$

which gives eq. (3.13).

Appendix C

The axial anomaly and the density of topological charge

The expansion for $\mathrm{tr}(F_{\mu\nu}{}^* F_{\mu\nu})$ for a self-dual solution, given in terms of the ADHM construction, contained in eqs. (4.22) and (4.23), can be verified directly and straightforwardly using the algebraic techniques of refs. [2–4]. On the other hand, this relation is just the way the equation connecting the divergence of the axial current to the density of the topological charge and the zero-mass solutions of the Dirac equation is realized for self-dual solutions. This was how we obtained the relation originally and we shall outline its proof along these lines.

We begin by using the ζ-function regularization procedure to give a discussion of the axial anomaly. The axial current is defined by suitably regularizing

$$J_\mu^5(x) = \mathrm{tr}(\gamma_\mu \gamma^5 S(x,y))\,, \tag{C.1}$$

where $S(x,y)$ is the fermion Green function defined by

$$\gamma \cdot DS(x,y) = -\delta(x-y) + \pi(x,y)\,, \tag{C.2}$$

and we follow the conventions of ref. [3] for γ-matrices, etc. In eq. (C.2), $\pi(x,y)$ is the projection operator on the zero-mass solutions of the Dirac equation:

$$\pi(x,y) = \sum_i \psi_i(x)\,\psi_i^+(y)\,, \tag{C.3}$$

where $\{\psi_i(x)\}$ is an orthonormal basis of solutions of

$$\gamma \cdot D\psi = 0\,. \tag{C.4}$$

We can calculate $S(x, y)$ given the Green function $G_F(x, y)$ defined by

$$(\gamma \cdot D)^2 G_F(x, y) = -\delta(x - y) + \pi(x, y)\,, \tag{C.5}$$

simply using

$$S(x, y) = \gamma \cdot DG_F(x, y) \tag{C.6}$$

If $\mathcal{G}_F(x, y; t)$ is the function $\mathcal{G}$ defined by eqs. (2.6) with $\mathcal{D} = -(\gamma D)^2$ and $\mathcal{G}_F$ the function obtained from $\mathcal{G}_F$ by projecting out the solutions of eq. (C.4),

$$\mathcal{G}'_F(x, y;t) = \mathcal{G}_F(x, y; t) - \int d^4z\ \mathcal{G}_F(x, z; t)\,\pi(z, y)\,, \tag{C.7}$$

we can define a regularized axial current by [31]

$$J^5_\mu(x) = \left\{\frac{1}{\Gamma(s)} \int_0^\infty dt\, t^{s-1}\ \mathrm{tr}[\gamma_\mu \gamma^5 \gamma \cdot D\,\mathcal{G}_F(x, y;t)]_{x=y}\right\}_{s=1}\,, \tag{C.8}$$

where the value at $s = 1$ is defined by analytic continuation from $\mathrm{Re}\, s > 2$. Then using

$$\vec{D}_\mu\ \mathcal{G}_F(x, y; t) = -\mathcal{G}_F(x, y; t)\,\overleftarrow{D}_\mu\,, \tag{C.9}$$

and $(\gamma \cdot D)^2\,\mathcal{G}_F = (\gamma \cdot D)^2\,\mathcal{G}'_F$ we obtain,

$$\partial_\mu J^5_\mu(x) = -\left\{\frac{2}{\Gamma(s)} \int_0^\infty dt\, t^{s-1}\ \mathrm{tr}[\gamma^5 (\gamma \cdot D)^2 \mathcal{G}'_F(x, y; t)]_{x=y}\right\}_{s=1} \tag{C.10}$$

$$= \left\{\frac{2(s-1)}{\Gamma(s)} \int_0^\infty dt\, t^{s-2}\ \mathrm{tr}[\gamma^5\,\mathcal{G}'_F(x, x; t)]\right\}_{s=1}\,. \tag{C.11}$$

So once again we have to evaluate the residue of a pole in an integral; this time we need the coefficient of the constant term in the asymptotic expansion of $\mathcal{G}'_F(x, y; t)$ as $t \downarrow 0$. For $\mathcal{G}_F(x, x; t)$

$$a_2(x, x) = \tfrac{1}{12} F_{\mu\nu}F_{\mu\nu} + \tfrac{1}{2}(\sigma_{\mu\nu}F_{\mu\nu})^2 + \tfrac{1}{6}D^2(\sigma_{\mu\nu}F_{\mu\nu})\,, \tag{C.12}$$

where $\sigma_{\mu\nu} = \frac{1}{4}[\gamma_\mu, \gamma_\nu]$, and

$$\mathrm{tr}(\gamma^5 a_2(x, x)) = \mathrm{tr}(F_{\mu\nu}{}^*F_{\mu\nu}) \tag{C.13}$$

From eq. (C.7) we see that the corresponding quantity for $\mathcal{G}'_F$, a'_2, satisfies

$$\mathrm{tr}(\gamma_5 a'_2(x, x)) = \mathrm{tr}(F_{\mu\nu}{}^*F_{\mu\nu}) - 16\pi^2\,\mathrm{tr}(\gamma^5 \pi(x, x))\,. \tag{C.14}$$

This yields the equation for the anomaly [31,32]

$$\partial_\mu J_\mu^5(x) = \frac{1}{8\pi^2}\operatorname{tr}(F_{\mu\nu}{}^*F_{\mu\nu}) - 2\sum_i \epsilon_i \operatorname{tr}(\psi_i(x)\,\psi_i^+(x))\,, \tag{C.15}$$

where $\gamma_5\psi_i = \epsilon_i\psi_i$. On integration eq. (C.15) yields the index theorem relating the topological charge k to the difference in the number of positive and negative chirality solutions of eq. (C.4).

If $F_{\mu\nu} = {}^*F_{\mu\nu}$ we can use the ADHM construction to obtain explicit expressions for the constituents of eq. (C.15). Then all the solutions to eq. (C.4) are of negative chirality, and [5]

$$\operatorname{tr}(\gamma^5\pi(x,x)) = -\frac{1}{\pi^2}\operatorname{tr}_W(fb^+Pbfb^+b) \tag{C.16}$$

$$\equiv \frac{1}{8\pi^2}\partial^2\operatorname{tr}_W(fb^+b)\,. \tag{C.17}$$

(To obtain eq. (C.17) we have used the result that

$$\operatorname{tr}_2(fb^+Pbf) = -\tfrac{1}{4}\partial^2 f\,, \tag{C.18}$$

where tr_2 indicates that the trace is only over the two-dimensional quaternion labels.)

To find an expression for $J_\mu^5(x)$ we note that for a self-dual solution $(\gamma\cdot D)^2$ $\frac{1}{2}(1+\gamma_5) = D^2$, and so

$$\gamma\cdot D\,\mathcal{G}_F(x,y;t) = \gamma\cdot\vec{D}\mathcal{G}(x,y;t)\,\tfrac{1}{2}(1+\gamma^5) + \mathcal{G}(x,y;t)\,\gamma\cdot\overleftarrow{D}\,\tfrac{1}{2}(1-\gamma^5)\,, \tag{C.19}$$

where $\mathcal{G}(x,y;t)$ is defined by eqs. (2.6) with $\mathcal{D} = -D^2$. Substituting in eq. (C.8),

$$J_\mu^5(x) = \left\{\frac{-2}{\Gamma(s)}\int_0^\infty \mathrm{d}t\, t^{s-1}\operatorname{tr}[\vec{D}_\mu\mathcal{G}(x,y;t) - \mathcal{G}(x,y;t)\overleftarrow{D}_\mu]_{x=y}\right\}_{s=1} \tag{C.20}$$

$$= \left\{-\frac{2(s-1)}{\Gamma(s)}\int_0^\infty \mathrm{d}t\, t^{s-2}\int \mathrm{d}^4z\,\operatorname{tr}[\vec{D}_\mu G(x,z)\,\mathcal{G}(z,y;t)\right.$$

$$\left. - \mathcal{G}(x,z;t)\,G(z,y)\overleftarrow{D}_\mu]_{x=y}\right\}_{s=1}\,. \tag{C.21}$$

This may be evaluated by the same techniques as eq. (2.30); again the singular part of G does not contribute, giving

$$J_\mu^5(x) = -2\operatorname{tr}[\vec{D}_\mu R(x,y) - R(x,y)\overleftarrow{D}_\mu]_{x=y} \tag{C.22}$$

$$= \frac{1}{4\pi^2}\operatorname{tr}[v(x)^+b\,\partial_\mu f(x)\,b^+v(x)] \tag{C.23}$$

$$= \frac{1}{8\pi^2} \partial_\mu \operatorname{tr}[Pbfb^+] , \tag{C.24}$$

where to obtain eq. (C.23) we have used eq. (3.13), and eq. (C.24) may be established using the properties of P, etc.

Substituting from eqs. (C.17) and (C.24) into eq. (C.15) we obtain eq. (4.22) with T given by eq. (4.23).

Appendix D

The matrix M and the conformal group

In sect. 4 we used the matrix M, which had previously occurred in the construction of the adjoint representation Green function [6], to construct a matrix function λ of the matrices a, b specifying the instanton solution having the properties of eqs. (4.31) and (4.23). Here we shall motivate the construction of M by considerations more directly related to the present analysis.

All the information necessary for the construction of the instanton solution specified by a, b is contained in the $2k$-dimensional hermitian quaternionic matrix [2,33]

$$N(a, b) = \begin{bmatrix} a^+a & a^+b \\ b^+a & b^+b \end{bmatrix} \tag{D.1}$$

$$\equiv \begin{bmatrix} \mu & \rho_\mu e_\mu \\ \rho_\mu e_\mu^+ & v \end{bmatrix} . \tag{D.2}$$

where the quadratic conditions on a and b are equivalent to the statement that the six $k \times k$ matrices μ, ν, ρ_μ, defined by eq. (D.2), are real and symmetric. Given N, a and b are specified up to a transformation of the form $a \to Qa$, $b \to Qb$, $Q \in \mathrm{Sp}(n + k)$, although conditions on the rank of N must be satisfied if it is to yield a solution for $n < k$. The transformation (3.8) induces

$$N \to (k^{\mathrm{T}} \oplus k^{\mathrm{T}}) N (k \oplus k) . \tag{D.3}$$

The transformation

$$a \to a' = a\chi + b\beta , \tag{D.4a}$$

$$b \to b' = a\gamma + b\alpha , \tag{D.4b}$$

induced by the conformal transformation (4.1), in turn induces a linear transformation on μ_{ij}, ν_{ij}, $\rho_{\mu ij}$, for each fixed i,j, defined by

$$h_{ij} \to h'_{ij} = s^+ h_{ij} s , \tag{D.5}$$

where

$$h_{ij} = \begin{bmatrix} \mu_{ij} & \rho_{\mu ij} e_\mu \\ \rho_{\mu ij} e_\mu^+ & \nu_{ij} \end{bmatrix}, \qquad s = \begin{bmatrix} \chi & \gamma \\ \beta & \alpha \end{bmatrix}. \tag{D.6}$$

It is convenient to define

$$\Gamma_\mu = \begin{pmatrix} 0 & e_\mu \\ e_\mu^+ & 0 \end{pmatrix}, \qquad \mu = 1 \dots 4 ; \qquad \Gamma_5 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \qquad \Gamma_6 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{D.7}$$

and also $\widetilde{\Gamma}_p = \Gamma_p$ for $p = 1, \dots, 5$ but $\widetilde{\Gamma}_6 = -\Gamma_6$. Then

$$\Gamma_p \widetilde{\Gamma}_q = g_{pq} + \Sigma_{pq} , \tag{D.8}$$

where $g_{pq} = 0$ for $p \neq q$, $q_{pp} = 1$ for $p = 1, g_{66} = 1$, and $\Sigma_{pq} = -\Sigma_{qp}$. Further

$$\widetilde{\Gamma}_b \Gamma_q = g_{pq} + \widetilde{\Sigma}_{pq} \tag{D.9}$$

with $\widetilde{\Sigma}_{pq} = -\Sigma_{pq}^+$. Both Σ_{pq} and $\widetilde{\Sigma}_{pq}$ obey the Lie algebra of O(5,1):

$$[\Sigma_{pq}, \Sigma_{rs}] = 2(g_{qr}\Sigma_{ps} - g_{pr}\Sigma_{qs} - g_{qs}\Sigma_{pr} + g_{ps}\Sigma_{qr}). \tag{D.10}$$

Using this formalism,

$$h_{ij} = h_{ij}^p \Gamma_p , \tag{D.11}$$

with an implied summation from $p = 1$ to 6. For an infinitesimal conformal transformation we may write

$$s = 1 - \tfrac{1}{4}\delta\omega^{pq}\widetilde{\Sigma}_{pq} + \tfrac{1}{2}\delta\kappa , \tag{D.12}$$

where $\kappa = \det s$, as in sect. 4. Then, since

$$\Sigma_{pq}\Gamma_r - \Gamma_r\Sigma_{pq} = 2(g_{qr}\Gamma_p - g_{pr}\Gamma_q) , \tag{D.13}$$

$$\delta h_{ij}^p = \delta\omega^{pq} g_{qr} h_{ij}^r + \delta\kappa h_{ij}^p . \tag{D.14}$$

Thus for each i,j h_{ij} defines a six-vector under the O(5,1) group generated by the $\{\Sigma_{pq}\}$. To satisfy condition (4.32), λ must be invariant under this O(5,1) group. Any such invariants may be expressed in terms of scalar products:

$$-2g_{pq}h_{ii'}^p h_{jj'}^q = \mu_{ii'}\nu_{jj'} + \mu_{jj'}\nu_{ii'} - 2\rho_{\mu ii'}\rho_{\mu jj'} , \tag{D.15}$$

except for those formed from the six-dimensional ϵ symbol. But even in the latter case the squares of these invariants are expressible in terms of scalar products so that the invariants of eq. (D.15) are complete up to a sign. The right-hand side of eq.

(D.15) defines the matrix M^{-1} introduced in ref. [6] and used in sect. 4.

We note further that, if M_S^{-1} and M_A^{-1} are defined from M^{-1} by symmetrising or antisymmetrising, respectively, on both sets of indices, they transform irreducibly, in the same way as M^{-1}, under transformation (3.8), but have dimensions $\frac{1}{2}k(k+1)$ and $\frac{1}{2}k(k-1)$, respectively. So (4.36) is replaced by

$$\det M_S^{-1} \to \det M_S^{-1} (\det \kappa)^{2k(k+1)} , \tag{D.16a}$$

$$\det M_A^{-1} \to \det M_A^{-1} (\det \kappa)^{2k(k-1)} . \tag{D.16b}$$

From eqs. (D.16) we deduce that

$$(\det M^{-1})^{k+1}/(\det M_S^{-1})^{2k} \tag{D.17}$$

is invariant under (3.8) and it is a conformal invariant. For the special case of the 't Hooft solutions discussed in sect. 5,

$$\det M^{-1} = \det M_S^{-1} \det M_A^{-1} , \tag{D.18}$$

and the techniques of sect. 5 may be applied to show

$$\det M_S^{-1} = 2^k \lambda_0^{2(k^2-1)} \Big(\prod_{i=0}^{k} \lambda_i^2\Big)^{k+1} \prod_{0\leqslant i<j\leqslant k} s_{ij} , \tag{D.19}$$

and the conformal invariant of eq. (D.17) is proportional to

$$(\det p)^{k+1} \prod_{0\leqslant i<j\leqslant k} s_{ij}^2 . \tag{D.20}$$

It may be used to show the equivalence of our results on the conformal transformation properties of $\det(-D^2)$ to Yoneya's [19], and gives the same conformal invariant as obtained by Schwarz [20] for $k = 2$.

References

[1] M.F. Atiyah, N.J. Hitchin, V.G. Drinfeld and Yu. I. Manin, Phys. Lett. 65A (1978) 185.
[2] V.G. Drinfeld and Yu. I. Manin, Comm. Math. Phys. 63 (1978) 177.
[3] E. Corrigan, D.B. Fairlie, P. Goddard and S. Templeton, Nucl. Phys. B140 (1978) 31.
[4] N.H. Christ, E. Weinberg, and N.K. Stanton, Phys. Rev. D18 (1978) 2013.
[5] H. Osborn, Nucl. Phys. B140 (1978) 45.
[6] E. Corrigan, P. Goddard and S. Templeton, Nucl. Phys. B151 (1979) 93.
[7] G. 't Hooft, Phys. Rev. Lett. 37 (1976) 8; Phys. Rev. D14 (1976) 343; (E:D18 (1978) 2199).
[8] C.G. Callan, R. Dashen and D.J. Gross, Phys. Rev. D17 (1978) 2217;
D.J. Gross, Phys. Reports 49 (1979) 143;
S. Coleman, The uses of instantons, Lectures at Erice (1977).
[9] M.F. Atiyah and D. Jones, Comm. Math. Phys. 61 (1978) 97.
[10] I.V. Frolov and A.S. Schwarz JETP Lett. 28 (1978) 273.
[11] B. Berg and M. Lüscher, DESY preprint 79/17.

[12] E. Witten, Nucl. Phys. B149 (1979) 285.
[13] D.B. Ray and I.M. Singer, Adv. Math. 7 (1971) 145.
[14] J.S. Dowker and R. Critchley, Phys. Rev. D13 (1976) 3224; Phys. Rev. D16 (1977) 3390
[15] S. Hawking, Comm. Math. Phys. 55 (1977) 133.
[16] S. Chadha, A. d'Adda, P. di Vecchia and F. Nicodemi, Phys. Lett. 72B (1977) 103;
F. T. Ore, Phys. Rev. D16 (1977) 2577;
A.A. Belavin and A.M. Polyakov, Nucl. Phys. B123 (1977) 429;
Yu. A. Bashilov and P. Pokrovsky, Nucl. Phys. B143 (1978) 431;
G.M. Shore, Cambridge preprint DAMTP 78/19; Ann. Phys., to appear.
C. Bernard, UCLA preprint (January, 1979).
[17] L.S. Brown and D.B. Creamer, Phys. Rev. D18 (1978) 3695.
[18] A. d'Adda and P. di Vecchia, Phys. Lett. 93B (1978) 162.
[19] T. Yoneya, Phys. Lett. 71B (1977) 407.
[20] A.S. Schwarz, Lett. Math. Phys. 2 (1978) 201, 247; Comm. Math. Phys. 64 (1978) 233;
I.V. Frolov and A.S. Schwarz, Phys. Lett. 80B (1978) 406.
[21] G. 't Hooft, unpublished;
R. Jackiw, C. Nohl and C. Rebbi, Phys. Rev. D15 (1977) 1642.
[22] P.B. Gilkey, The index theorem and the heat equation (Publish or Perish, 1974).
[23] S. Minakshisundaram and A. Pleijel, Can. J. Math. 1 (1949) 320;
R.T. Seeley, Proc. Symp. Pure Math. 10 (1971) 288.
[24] B.S. deWitt, Dynamical theory of groups and fields (Gordon and Breach, 1965); Phys. Reports 19 (1975) 295.
[25] H.P. McKean and I.M. Singer, J. Diff. Geom. 5 (1971) 233;
P.B. Gilkey, J. Diff. Geom. 10 (1975) 601; Proc. Symp. Pure Math. 27 (1975) 265;
N.K. Nielsen, NORDITA preprint 78/24.
[26] E. Corrigan and B. Hasslacher, Phys. Lett. 81B (1979) 181.
[27] L.S. Brown, R.D. Carlitz, D.B. Creamer and C. Lee, Phys. Rev. D17 (1978) 1583.
[28] J.J. Sylvester, Quarterly, J. Math. 1 (1855) 42.
[29] C.W. Borchardt, Crelle's J. 57 (1859) 111.
[30] A. Patrasciou, University of Arizona preprints.
[31] J.S. Dowker, J. Phys. A11 (1978) 347.
[32] N.K. Nielsen and B. Schroer, Nucl. Phys. B127 (1977) 493.
[33] C. Meyers and M. de Roo, Nucl. Phys. B148 (1979) 61.
[34] A.A. Belavin, V.A. Fateev, A.S. Schwarz and Yu. S. Tyupkin, Phys. Lett. 83B (1979) 317.